GOORRT: The Governed Object-Operation-Realization Representation Triad

A Jurisdictional Representation Architecture for Governed Semantic Computing

GOML addresses governed object and state meaning; GSOL addresses governed operations and transitions; MCIR addresses machine-oriented realization representations.

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Abstract

Governed semantic systems repeatedly collapse three different questions: what an object or state means, what operations may change it, and how a selected operation is computationally realized. GOORRT proposes a three-family representation architecture to keep those jurisdictions distinct while connecting them through typed correspondence, authority, evidence, gap, refusal, and standing records. GOML addresses governed object and state meaning; GSOL addresses governed operations and transitions; MCIR addresses machine-oriented realization representations. The triad is best treated as a candidate primary projection for the current corpus, not as a proof that every representation concern reduces to three axes. A representation is not the object, operation, execution, authority, or evidence it denotes. The paper identifies native-jurisdiction, projection, and co-representation cases; locates supporting records without promoting them automatically into peer languages; maps GOORRT into the universal Governed Semantic Model and the Possibility-Specification-CM1-CM2-IC-IR-Evidence-Revision funnel; and states counterexamples that would require extension or demotion.

Keywords: governed representation; object models; operation contracts; intermediate representation; correspondence; semantic projection; Constitutional Computing; GMB; universal GSM.

Purpose

GOORRT – the **Governed Object–Operation–Realization Representation Triad** – names three primary representation families proposed for a constitutionally governed semantic/computational substrate:

1. **GOML** – Governed Object/Model Language
2. **GSOL** – Governed Semantic Operation Language
3. **MCIR** – Meta-Compilation Intermediate Representation

Together, these form the core representational scaffold for Jnana, MCIR migration, MCRM comparison, and later MosaiCOS work.

GOORRT exists to prevent representation collapse. It keeps object semantics, operation semantics, and computational realization distinct while allowing them to be bound by explicit correspondence, admission, receipt, gap, and refusal records.

The slogan form:

```
GOML says what object or state is represented.
GSOL says what operation or transition is represented as admissible,
refused, or observed.
MCIR says how a computational realization is represented.
```

Or, more formally:

```
GOML = governed object/state representation
GSOL = governed operation/transition representation
MCIR = governed computational realization representation
```

Core Claim

GOORRT is a **candidate primary representation triad** for the current GOML/GSOL/MCIR/Jnana architecture. It proposes that three recurring jurisdictions should remain distinct:

1. governed object and state meaning;
2. governed operations and transitions;
3. computational realization representation.

The claim is architectural and falsifiable, not a completeness theorem. A fourth peer family is unnecessary only when an additional concern can be represented without loss as a typed profile, relation, evidence object, context, or bridge record attached to one or more of the three families. Authority, provenance, time, intent, evidence, uncertainty, embodiment, interaction, and institution remain possible counterexamples. If any requires independent syntax, semantics, validity rules, evolution, and irreducible transformations that the triad cannot carry without distortion, GOORRT must expand or narrow its scope.

GOORRT therefore prevents a useful class of representation collapse without claiming that three files, languages, or schemas exhaust semantic reality.

The Three Representation Families

1. GOML – Governed Object/Model Language

GOML is proposed as the governing representation family for governed objects, states, institutions, roles, authorities, identities, and semantic structures.

GOML answers:

```
What is being represented?
What object, institution, role, state, or standing exists?
What boundaries and identity conditions define it?
What jurisdiction or authority profile applies?
```

Examples of GOML-native structures include:

```
object model
institution model
role model
authority model
jurisdiction model
standing state
epistemic position
semantic object boundary
```

GOML may be native for a claim when the thing being represented is primarily semantic, institutional, legal, constitutional, or authority-bearing. In those cases, MCIR may serve as a computational projection, but not as the native home of meaning.

2. GSOL – Governed Semantic Operation Language

GSOL is proposed as the governing representation family for governed operations, transitions, effects, refusals, gaps, receipts, standing ceilings, and admission rules.

GSOL answers:

```
What operation is being proposed?
What transition is allowed?
What effects are permitted or forbidden?
What receipts are required?
What gaps or refusals must be recorded?
What standing may this operation earn?
```

Examples of GSOL-native structures include:

```
operation contract
admission operation
refusal operation
gap certificate
transition rule
```

```
receipt requirement
standing ceiling
nonclaim ledger
anti-loop rule
blocker triage rule
```

GSOL may be native for a claim when the core meaning lies not in an object or program, but in the governed act itself: an admission, refusal, delegation, migration, comparison, rebuild, or standing transition.

3. MCIR – Meta-Compilation Intermediate Representation

MCIR is proposed as the governing representation family for machine-oriented descriptions of computational realization, executable structure, compilation, replay, transformation, and checkable behavior.

MCIR answers:

```
How is this computationally realized?
How can it execute, compile, replay, transform, or compare?
What identified computational form is controlling under the selected
profile?
What receipts or traces does execution produce?
```

Examples of MCIR-native structures include:

```
program representation
intermediate representation
compilation unit
execution graph
replay form
transformation pipeline
binary/source relationship surface
computational witness
```

MCIR may be native for a claim when the thing being represented is fundamentally computational. In such cases, GOML and GSOL describe the object and governance structure around it, while MCIR carries the identified machine-oriented realization representation.

GOORRT Is Not a Hierarchy

GOORRT should not be modeled as:

```
GOML -> GSOL -> MCIR
```

or:

```
MCIR as the only real canonical representation
```

Instead, the relationship is a governed correspondence loop:

```
GOML object/state
  is acted on by
GSOL operation/transition
  which may be realized or checked through
MCIR computational form
  which emits receipts/traces/gaps
  which update
GOML standing/state
```

The loop form is important because different cases have different native homes.

Three Correspondence Cases

Case 1 – GOML/GSOL as Source of Truth

The object or operation is primarily semantic, institutional, legal, constitutional, or governance-native.

In this case:

```
GOML/GSOL are native for the identified semantic or operational claim.
MCIR is a computational projection or checking realization.
```

Example:

```
A jurisdictional standing change is not reducible to MCIR.
It can be checked or carried by MCIR, but its native meaning is
GOML/GSOL.
```

Case 2 – MCIR as Source of Truth

The object is fundamentally computational.

In this case:

```
MCIR is native for the identified computational claim.
GOML/GSOL describe the object boundary, operation contract, and
governance conditions.
```

Example:

```
A compiled fixture or transformation pipeline may treat an
identified MCIR artifact as the controlling representation for a
bounded computational claim.
```

Case 3 – Co-Represented Claims

Neither side fully reduces to the other.

In this case:

```
GOML, GSOL, and MCIR are bound by an explicit
correspondence/admission record.
```

This is the expected default for Stage B MCRM/Jnana migration work.

Supporting Profiles and Binding Records

The primary triad is complete, but Stage B requires typed supporting records. These should not be treated as fourth, fifth, or sixth peer languages. They are profiles or bridge records inside the GOORRT architecture.

JAR – Jurisdiction / Authority Representation

Mostly a GOML profile.

Represents:

```
jurisdiction
authority
role
standing ceiling
admission authority
constitutional profile
```

EPR – Epistemic Position Representation

Mostly a GOML profile, updated by GSOL operations.

Represents:

```
what is currently admitted
what standing has changed
what the system may believe after receipts
what remains unadmitted
```

GRR – Gap / Refusal Representation

Mostly a GSOL profile.

Represents:

```
missing proof
refused claim
blocked transition
nonclaim
gap certificate
condition required for future admission
```

RTR – Receipt / Trace Representation

An evidence schema produced by GSOL/MCIR operations.

Represents:

```
inputs
outputs
hashes
commands
environment
execution traces
build transcripts
comparison receipts
```

CAR – Correspondence / Admission Record

A bridge record binding GOML, GSOL, MCIR, MCRM, receipts, gaps, and epistemic state.

CAR is the most important Stage B bridge.

It represents:

```
this GOML object
under this GSOL operation
corresponds to this MCIR realization
under this MCRM comparison
with these receipts
with these gaps
yielding this admission or refusal
```

CAR prevents the system from collapsing correspondence into equivalence.

Stage B Use

In Stage B, GOORRT should be used to structure every MCRM/Jnana migration fixture.

A fixture should answer:

```
GOML:
  What object, model, state, or fixture is being represented?

GSOL:
  What governed operation is being attempted?

MCIR:
  What computational realization is being produced, checked, executed,
  or compared?
```

```
JAR:
  What authority and jurisdiction make the operation admissible?

CAR:
  What correspondence is claimed among GOML, GSOL, MCIR, and MCRM?

RTR:
  What witnessed transition do the receipts support?

GRR:
  What gaps, refusals, or nonclaims remain?

EPR:
  What standing changes after admission?
```

This gives Stage B a precise anti-overclaim structure.

Forbidden Collapses

GOORRT exists partly to forbid the following collapses:

```
GOML = MCIR
GSOL = MCIR
MCRM comparison = equivalence
fixture rehearsal = migration complete
MCIR candidate = MCIR admitted
bounded correspondence = full semantic identity
receipt emitted = standing earned
operation executed = authority granted
```

Any claimed correspondence should be declared, evidenced, bounded, and separately admitted where admission is required.

Stage B Claim Discipline

GOORRT supports these allowed claims:

```
fixture selected
representation bound
MCIR candidate identified
Jnana realization rehearsed
MCRM comparison recorded
receipt admissibility witnessed
fixture-scoped admission made
gap/refusal recorded
standing updated under profile
```

GOORRT forbids premature claims such as:

```
equivalence
full migration
self-hosting
bootstrap complete
production runtime
MosaiCOS started
trusted compiler
Trusting Trust resolved
```

Relationship to MCRM

MCRM is not a fourth representation family. It is a comparison method or comparison regime used to relate representations without collapsing them.

Within GOORRT:

```
GOML defines the object/state.
GSOL defines the operation.
MCIR defines the computational realization.
MCRM compares witnessed realizations.
CAR records the correspondence/admission result.
RTR supplies the receipts.
GRR records gaps/refusals.
EPR records the updated standing.
```

MCRM should never emit unqualified equivalence. Its strongest honest claim is bounded correspondence under declared inputs, receipts, and gaps.

Why Three Families May Be Sufficient

The triad covers three high-value questions for governed computation:

```
Object / State: What is represented as existing or holding?
Operation / Transition: What change is proposed, admitted, refused,
or observed?
Realization Representation: By what machine-oriented structure is a
selected computation described or carried?
```

Supporting concerns remain necessary. JAR, EPR, GRR, RTR, and CAR are candidate profiles or bridge records only if their own semantics can be expressed compositionally and queried without erasing their jurisdiction. The classification cannot be decided by naming convention. A receipt schema may be attached to an operation or realization while still requiring a distinct evidence ontology. An authority record may profile an object while authority validation remains a separate operation. “Not a peer language” is therefore a design proposal to test, not a status that GOORRT can grant by assertion.

The strongest present claim is: the triad is a compact organizing projection that exposes object/operation/realization confusion across the reviewed sources. Its sufficiency remains contingent on schemas, examples, loss analysis, round-trip behavior, and competing decompositions.

Recommended Bounded Definition

GOORRT is the **Governed Object-Operation-Realization Representation Triad**: a candidate three-family architecture for keeping governed object/state meaning, governed operation/transition meaning, and machine-oriented realization representation distinct while binding them through explicit, versioned, and evidence-bearing correspondence relations.

In this bounded definition:

```
GOML represents governed objects, states, roles, institutions, and
semantic structures.
GSOL represents governed operations, transitions, effects, refusals,
and obligations.
MCIR represents machine-oriented realization, compilation,
transformation, replay, or comparison structures.
```

Each family may be native for a claim without becoming the only real representation. A representation's canonical identity, if granted by an applicable profile, does not make it identical to its referent or guarantee that a parser, runtime, or effect conforms to it. Correspondence, translation, admission, and evidence remain separate relations.

Integration Note for GOML/GSOL Work

The GOML/GSOL session should treat GOORRT as the formal representation scaffold for Stage B.

GOML and GSOL should not be presented merely as projections of MCIR. They may be native for claims within their own jurisdictions.

MCIR may be described as the controlling computational realization representation under an applicable profile, not as the only real representation.

Stage B should introduce CAR as the fixture-level binding record that connects GOML, GSOL, MCIR, and MCRM.

A future Stage B fixture proposal would need to identify:

```
GOML object record
GSOL operation contract
MCIR candidate or MCIR gap certificate
MCRM comparison policy
CAR correspondence/admission record
RTR receipt bundle
```

```
GRR gap/refusal/nonclaim ledger
EPR standing update
```

This makes GOORRT operational rather than merely descriptive.

Representation Is Not Reality

GOORRT names representation families. GOML is not the object or institution; GSOL is not the operation or authority to perform it; MCIR is not the running system or effect. Each representation requires an exact identity, schema/version, interpretation environment, and claim boundary. A correspondence record must identify the source and target representations, relation type, translation or comparison procedure, preservation obligations, known loss, evidence, authority basis, and standing ceiling.

The three source cases are clearer when phrased as **native jurisdiction** rather than unqualified "source of truth":

- In a semantic-native case, GOML or GSOL controls the intended object or operation meaning, while MCIR is a projection or checking representation.
- In a computation-native case, MCIR controls the identified computational structure, while GOML and GSOL describe its semantic boundary and governed use.
- In a co-representation case, no family is globally controlling; a typed correspondence contract states which properties must agree and which differences are permitted.

Co-representation is not co-identity. Round-trip stability, observational equivalence, refinement, simulation, trace correspondence, and bounded semantic preservation are different claim types and require different evidence.

GMB, Universal GSM, and the Realization Funnel

The **Governed Meaning Base (GMB)** is the current name in the author-confirmed `CCF -> CMB -> GMB` lineage. GMB is a governed meaning and lineage substrate in which GOML, GSOL, MCIR, their schemas, relationships, receipts, gaps, standing, and revision histories may be stored and queried. GOORRT is a representation pattern within that wider fabric, not a replacement for GMB and not proof that stored meanings correspond to reality.

Within the universal **Governed Semantic Model (GSM)**, the triad can project across the realization funnel:

$$
\begin{aligned}
& P \\
& \to S \\
& \to CM_1 \\
& \to CM_2 \\
& \to IC \\
& \to IR \\
& \to E \\
& \to R
\end{aligned}
.
$$

Possibility contains candidate objects, operations, representations, translations, and realizations. **Specification** defines their types, invariants, semantics, and correspondence obligations. **Configuration Manifold 1 (CM1)** contains all admissible representation profiles, schemas, translators, and realization choices. **Configuration Manifold 2 (CM2)** selects preferred admissible choices under declared objectives such as fidelity, inspectability, cost, portability, and assurance. **IC** means **Instance Constitution** or **Instance Configuration** interchangeably and binds the selected schema versions, relations, authorities, and evidence profile. **IR** means **Instance Realization** or **Implementation Realization** interchangeably and contains the actual documents, graphs, code, binaries, runtimes, and effects. **Evidence** evaluates syntax, semantics, correspondence, execution, and observed effects. **Revision** changes any earlier layer while retaining lineage.

No family maps one-to-one onto one funnel stage. GOML, GSOL, and MCIR can all appear in Specification, both manifolds, IC, IR descriptions, evidence, and revision records. The funnel concerns realization and standing; GOORRT concerns representation jurisdiction.

Primitive and Invariant Kernel

The integrated primitive set is: **representation artifact**, **representation family**, **schema/version**, **referent**, **native jurisdiction**, **projection**, **translation edge**, **correspondence contract**, **preservation obligation**, **loss/gap declaration**, **comparison method**, **evidence bundle**, **authority basis**, **standing ceiling**, and **revision lineage**.

The governing invariants are:

1. A representation is not identical to its referent.
2. Execution evidence does not grant execution authority or representational canonicity.
3. Syntactic validity does not establish semantic correspondence.
4. Correspondence does not imply equivalence; equivalence does not imply identity.

5. A translation must state preserved, transformed, omitted, and unknown properties.
6. A bridge record cannot award itself standing.
7. A family declared native for one claim does not control every claim about the artifact.
8. Schema, interpreter, comparator, and runtime identities belong to the evidence boundary.
9. Supporting profiles remain distinct even when encoded inside a shared carrier.
10. Revision must identify invalidated correspondence and dependent claims.

Jurisdictional Representation Decomposition

The completed P1 corpus sharpens GOORRT into a **jurisdictional decomposition of object/model, operation, and realization representations with explicit correspondence records**. The triad prevents one representation family from silently governing questions outside its jurisdiction. It is not a hierarchy, a complete language standard, or proof that exactly three families suffice in every domain.

- **Adjacent work:** *The Ableman Format Specification (AFS)*
  Supplies: Format families, profiles, schemas, and conformance claims.
  GOORRT boundary: GOORRT is one representational architecture that AFS may profile.

- **Adjacent work:** *Representational Topology in MCIR: AST, DAG, and Emergent Governance Capabilities*
  Supplies: Separation of authoring AST, canonical DAG, and receipt graph.
  GOORRT boundary: That paper governs internal topology; GOORRT governs jurisdiction among representation families.

- **Adjacent work:** *Governed Controlled Natural Language (GCNL)*
  Supplies: Human-authorable proposal syntax.
  GOORRT boundary: GCNL may express object or operation proposals before admission; it is not one of the triad by identity.

- **Adjacent work:** *Governed Controlled Machine Language (GCML)*
  Supplies: A machine-facing procedural profile.
  GOORRT boundary: GCML may realize portions of GSOL; exact identity and correspondence require a profile decision.

- **Adjacent work:** *Governance as Compilation*
  Supplies: Parsing, translation, admission, lowering, residue, and correspondence.
  GOORRT boundary: GOORRT supplies typed representation domains consumed by that process.

A co-representation case should bind:

$$R = (object_representation, operation_representation, realization_representation, jurisdictions, source_of_truth, correspondence, residuals, evidence, standing, revision)$$

GOML describes governed objects or models. GSOL describes governed operations or transitions. MCIR carries canonical machine-legible commitments across transformations. A domain may designate one source of truth, use co-representation, or adopt another decomposition. Every choice must state what happens when representations disagree.

Correspondence is not equality. A valid object model can have several admissible operation sets; one operation can have several realizations; a realization can diverge from both model and operation; and evidence can defeat a previously admitted correspondence. JAR, EPR, GRR, RTR, and CAR remain supporting records rather than new sources of authority by name.

The universal GSM can type
`REPRESENTS_GOVERNED_OBJECT`, `REPRESENTS_GOVERNED_OPERATION`,
`REPRESENTS_REALIZATION_COMMITMENT`, `HAS_REPRESENTATION_JURISDICTION`,
`DESIGNATED_SOURCE_OF_TRUTH`, `CORRESPONDS_UNDER_CAR`, `DISAGREES_WITH_REPRESENTATION`,
`LEAVES_REPRESENTATIONAL_RESIDUAL`,
and
`REVISED_AFTER_DIVERGENCE`. This makes competing decompositions comparable.

This integration establishes four editorial commitments: adopt jurisdictional representation decomposition as GOORRT's official corpus role; adopt the co-representation envelope; require explicit disagreement and source-of-truth policies; and retain triad sufficiency as a defeasible design hypothesis rather than a universal primitive count.

Representation-to-Effect Jurisdiction Boundary

The fixed-seed comparison with SMT retrieval, Meta-Binding, communal computing, and APD clarifies that co-representation ends before physical or institutional effect unless a separately governed correspondence is supplied:

$$J = (object_representation, operation_representation, realization_representation, correspondence, effect_surface, authority, evidence, remainder)$$

SMT can admit nodes and paths from several representation families while refusing or leaving unknown their correspondence. Meta-Binding can test whether a represented operation over object constraints changes later behavior, but that

functional result does not prove that the realization conforms or that the effect was contained.

Communal computing can project governed objects, proposed operations, and realized public artifacts into shared physical space. Placement, gesture, possession, or majority attention does not establish source-of-truth status or correspondence. Each public interpretation needs a declared representation family and appealable mapping.

APD applies at the realization and effect boundary. A complete GOORRT triad can describe a containment plan while leaving bypasses, hidden channels, provider effects, or irreversible remainder outside the representation. Representation conformance is therefore not an APD claim.

The universal GSM can type
`ADMITS_CROSS_REPRESENTATION_PATH`,
`TESTS_OPERATION_OVER_OBJECT_MODEL`, `PROJECTS_REPRESENTATION_IN_CIVIC_SPACE`,
`CORRESPONDS_TO_REALIZATION_UNDER_EVIDENCE`, `ENTERS_APD_EFFECT_SURFACE`,
and
`DOES_NOT_PROVE_EFFECT_CONTAINMENT`.

This randomized review establishes four editorial commitments: add the effect surface and authority boundary to co-representation cases; keep retrieval admission below correspondence; make communal projections representation-explicit; and prohibit representational completeness from implying APD containment.

Claim-Relative Representation Jurisdiction

Ten downstream uses apply GOORRT to communal objects and shared operations, agentic courts, controlled-language object models, operational constraints, retrieval, legacy migration, binding-quality evaluation, sovereignty, enforcement, and APD containment. They confirm the value of separating object/model, operation, and realization representations while also exposing the danger of treating the triad as complete by declaration.

The correspondence case is:

$$\begin{aligned} C^* = (&claim, families, native_jurisdiction, source_of_truth,\\ &representations, correspondences, translations,\\ &preservation, loss, evidence, authority, standing,\\ &disagreement, invalidation, effect_boundary, remainder) \end{aligned}.$$

Source of truth is claim-relative. GOML may govern one object-state claim, GSOL one operation contract, and MCIR one machine-oriented realization description. Co-representation requires an explicit disagreement policy; no file, graph, or compiler output can silently become native authority for every claim.

JAR, EPR, GRR, RTR, and CAR remain candidate profiles or bridge records unless independent semantics justify peer-family standing.

Representation completeness is not effect closure. A full triad can omit bypasses, actors, evidence ontologies, time, interaction, institutional law, provider behavior, or irreversible external remainder. The architecture should be compared with a unified typed graph and alternate decompositions, with independent classification, translation-loss, round-trip, and multi-or-no-family fixtures.

The universal GSM can type
`ASSIGNS_NATIVE_REPRESENTATION_JURISDICTION`,
`CORRESPONDS_ACROSS_GOORRT_FAMILY`, `PRESERVES_CLAIM_IN_TRANSLATION`,
`DISAGREES_ACROSS_REPRESENTATION`, `INVALIDATES_CORRESPONDENCE`,
`REMAINS_UPSTREAM_OF_EFFECT`, and
`FAILS_TRIAD_SUFFICIENCY_TEST`.

This reverse-dependency review establishes four editorial commitments: adopt the claim-relative correspondence case; require disagreement and invalidation policies; retain supporting records as provisional profiles; and keep triad sufficiency defeasible through competing-decomposition tests.

Counterevidence and Competing Decompositions

GOORRT should be expanded, narrowed, or rejected if: (1) authority requires an irreducible peer language rather than a GOML profile plus GSOL validation; (2) evidence and provenance require independent semantics that cannot be expressed without distortion; (3) time, interaction, context, institution, or uncertainty repeatedly crosscuts all three families; (4) GOML and GSOL cannot be separated without duplicating most syntax and semantics; (5) MCIR varies so widely by runtime that it is a family label without stable obligations; (6) co-representation records cannot prevent semantic drift; (7) two independent modelers cannot classify the same artifacts consistently; (8) round-trip translations systematically lose properties the architecture calls preserved; (9) supporting records evolve as independent languages; (10) the triad encourages implementers to mistake encodings for effects; (11) the same result is achieved more simply by a typed graph without three named families; (12) the three-family boundary impedes domain-specific representations; (13) MCRM comparisons cannot state useful relations short of equivalence; or (14) a claimed complete triad conceals excluded actors, evidence, or authority.

Evaluation should compare GOORRT with at least one unified typed-graph model and one alternative decomposition. Required tests include independent classification, schema validation, translation loss, round-trip behavior, claim-specific correspondence, dependency invalidation, and adversarial examples that fit multiple or no families.

Separate Representation Roles Without Forcing Every System Into Three Boxes

Public orientation

GOORRT distinguishes representations of objects, operations, and realizations so a system does not silently substitute what exists, what may be done, and how it is computationally enacted. The triad is a disciplined decomposition and comparison tool. It is not proof that every representation belongs exclusively to one family or that correspondence between families is lossless.

Adversarial tests

1. **Forced-triad attack.** A mixed or reflexive representation is assigned to one family merely to preserve the model. Multi-role records and justified alternate decompositions must remain possible.
2. **MCRM-layer-creep attack.** The correspondence relation becomes an unacknowledged fourth representation family or a universal semantic oracle. Its inputs, outputs, and authority must remain explicit.
3. **Correspondence-as-equivalence attack.** A mapping is treated as identity, substitutability, or behavioral equivalence without a declared property set and direction.
4. **Lowering-loss attack.** Object meaning or operation obligation disappears during realization, while syntactic success is reported as semantic preservation.
5. **Authority-propagation attack.** Permission attached to an object or operation is assumed to authorize every realization that can encode it.
6. **Incomplete-operation attack.** Preconditions, refusals, side effects, concurrency, exception behavior, or recovery semantics are absent from the operation representation.
7. **Representation-reality attack.** A complete representation is mistaken for the represented system, environment, person, or institution.
8. **Vocabulary-collision attack.** Realization-stage IR is confused with Intermediate Representation, causing a stage, artifact, or assurance claim to be silently renamed.

Claim and evidence ceiling

The paper proposes three representation jurisdictions plus explicit correspondence machinery. It does not prove ontological completeness, exclusive classification, semantic identity, lossless translation, executable correctness, or authority conservation. MCRM is a bounded relation among declared representations, not independent proof that the relation is adequate.

Claims about compiler design, formal semantics, interoperability, verification, cognition, institutional modeling, or domain portability require external comparison and exact worked realizations. Here IR in the realization funnel means

Instance Realization or Implementation Realization; IR in a named representation such as MCIR means Intermediate Representation. No external research was added in this pass.

Reader-facing consequence

The accessible question is simple: are we talking about the thing, the allowed transformation, or the machinery that performs it - and what evidence shows that meaning and obligation survived movement between those views?

Corpus Integration Sources

This pass used exact local heads for AFS (SHA-256 `E5B246C68B3A7AFADB44F4E368F0037E6B9970F30AF04B583FAF993D009D4496`), GSOL v0.1 (

`44F8689BF2FD50670CD25347FB2363491115CA7C049C2BBF71D7286A226DD1AC`), MCIR (
`0E1BF423FDEA88CD1A9BB4AB95AAD44D0062D87419B901DB79D1376A391049E8`), Jnana (
`B5238D73BA03320D0D331A3D40AEEAB4264F41EA0DF50DA0AF61F769307F109D`), *The Three Hash Rule* (
`03F60317CC9AFB36DDBCF9BF1C6D185611666CC4D9AB290BD4BCC17DF4579B7C`), and GMB semantic-manifold documentation (
`49A05A424F226A21C81D638727E1165DE1AD55AB1CBDC56EA65D67A1362A4B08`). Comparison custody creates no identity merger, adoption, completeness proof, or standing.

Short Form

```
GOML says what object or state is represented.
GSOL says what governed operation or transition is represented.
MCIR says how a computational realization is represented.
CAR states a bounded correspondence among identified representations.
Receipts record witnessed transitions under declared evidence.
Gaps and refusals preserve unsupported claims as unsupported.
Standing changes only through separately authorized admission.
```

The short form is pedagogical. The schemas, interpreters, relations, evidence, and authority boundaries carry the actual claim.

Realization Scope and Evidence Boundary

This successor makes the paper's stage boundary explicit. It narrows claims rather than manufacturing missing evidence: a stage named as context is not thereby claimed as implemented, validated, or complete.

- **Possibility Space.** Existing source passages use this stage only as bounded context or an obligation. The paper does not claim an independently validated contribution at this stage.
- **Revision.** Existing source passages use this stage only as bounded context or an obligation. The paper does not claim an independently validated contribution at this stage.
- **Publication form.** This work is retained as a standalone technical companion because its concrete migration method is useful independently; it remains subordinate to the architecture it explains.

Editorial Provenance and Standing

Source, Evidence, and Standing

This is an AI-prepared successor revision candidate derived from one exact author manuscript. The integration custody manifest is `work/paper-saturation-v1/evidence/goorrt/`

`integration-custody-manifest.v0.1.json`,
SHA-256
`D6ABD8D2A9434DA6613A6D3CE886B179108362F29C8E0E690C536141F6F0CCA4`.
The exact source remains unchanged. The GOORRT name, object/operation/realization separation, representation families, supporting record proposals, Stage B framing, forbidden collapses, and concise slogan are author material. This pass turns a session-oriented architecture note into a bounded paper candidate: it demotes an unsupported completeness claim to a falsifiable architecture hypothesis, separates representations from the realities they describe, places GOORRT in the GMB/universal GSM and full realization funnel, and adds counterevidence and open editorial questions. Those changes are AI editorial or synthetic contributions until adopted. The source's imperative Stage B language is historical or proposed architecture content, not an instruction executed by this review. The controlling GOML and MCRM specifications and the named Stage B fixture package were not resolved as exact comparison sources. No language, parser, schema, compiler, migration, comparison, admission, or runtime was implemented or replayed. No publication, canonical standing, identity merger, conformance assignment, or implementation authority is granted.

Corpus-Wide Reference Positioning

This paper argues that governed semantic systems repeatedly collapse three different questions: what an object or state means, what operations may change it, and how a selected operation is computationally realized. Its nearest external comparison fields are formal, provenance, semantic. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework

or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: Formal assurance remains relative to the encoded model, trusted base, and refinement chain; proof does not remove specification error or unmodeled environmental behavior. A complete provenance graph can faithfully record a false or unauthorized process; provenance is evidence about lineage, not automatic truth or legitimacy. Formal vocabularies and graph links make commitments explicit but cannot by themselves establish correct identity, shared interpretation, or lossless translation between contexts. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[**E1**] George C. Necula. "Proof-Carrying Code." *POPL '97*, 1997, pp. 106-119. DOI: 10.1145/263699.263712. https://doi.org/10.1145/263699.263712 **Relationship:** Predecessor for binding machine-checkable evidence to executable artifacts; the present paper may demand broader semantic or constitutional obligations.

[**E2**] Xavier Leroy. "Formal Verification of a Realistic Compiler." *Communications of the ACM* 52(7), 2009, pp. 107-115. DOI: 10.1145/1538788.1538814. https://doi.org/10.1145/1538788.1538814 **Relationship:** Predecessor verified-compilation result used to distinguish proven translation preservation from

broader governance claims.

[**E3**] Patrick Cousot and Radhia Cousot. "Abstract Interpretation: A Unified Lattice Model for Static Analysis of Programs by Construction or Approximation of Fixpoints." *POPL '77*, 1977. DOI: 10.1145/512950.512973. https://doi.org/10.1145/512950.512973 **Relationship:** Predecessor formal framework for sound approximation and fixpoint reasoning.

[**E4**] Leslie Lamport. *Specifying Systems: The TLA+ Language and Tools for Hardware and Software Engineers.* Addison-Wesley, 2002. https://lamport.azurewebsites.net/tla/book.html **Relationship:** Predecessor specification discipline for state machines, safety, liveness, and refinement.

[**E5**] W3C. *PROV-O: The PROV Ontology.* W3C Recommendation, 30 April 2013. https://www.w3.org/TR/prov-o/ **Relationship:** Adjacent provenance standard; it models entities, activities, and agents, but does not by itself establish the paper's stronger authority or standing claims.

[**E6**] Santiago Torres-Arias et al. "in-toto: Providing Farm-to-Table Guarantees for Bits and Bytes." *28th USENIX Security Symposium*, 2019, pp. 1393-1410. usenix.org source **Relationship:** Predecessor supply-chain lineage system used for comparison; no influence claim is made unless the source head already documents one.

[**E7**] OpenSSF. *Supply-chain Levels for Software Artifacts (SLSA), Version 1.2.* 2026. https://slsa.dev/spec/v1.2/ **Relationship:** Adjacent software-supply-chain assurance standard, especially relevant to provenance and verification boundaries.

Ableman corpus

[**A1**] Adam Ableman. *MCIR: A Canonical Intermediate Representation for Governed Transformation.* Ableman Corpus current edition; work identity `PAPER.PUB.21095303`;
source SHA-256
`EA129ECBA95F7B07CAA7B33A54F91F52913C94A14DA227EB9A06CFD083E4E5BF`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *Governed Semantic Compilation.* Ableman Corpus current edition; work identity `PAPER.PUB.18453681`;
source SHA-256
`5FF6C636550B04DB0E2F6F94C379BB75AA0BD8781E19B047D7D3BB6D49071AC2`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A3**] Adam Ableman. *Hybrid Semantic Compiler Architectures.* Ableman Corpus current edition; work identity `PAPER.PUB.21228038`;
source SHA-256

EBCD6DDDA9B547639AEBED79FB68069473CF1248E41C527B48C9624BBE2C0496.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A4**] Adam Ableman. *Admissibility Engineering.* Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-F1E33B5E973B0900;
source SHA-256
CDC80271268802CA5CB4DE3624B992FB2FF07D677BF945C4B349BD307C4C2A6E.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A5**] Adam Ableman. *Everything Is a Compiler.* Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-30FB8F95BD990A47;
source SHA-256
FE0F88A1F125317285BE4378E5F27ACC9FEB83ADE3FF329A3DA9F9CB026DF9A9.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A6**] Adam Ableman. *Reasoning by Way of Receipts.* Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-04E427F6F4D94454;
source SHA-256
AF86A66CCD6A103DD75208500AE99148911A6453B6DFDF36A954279F21AB8489.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A7**] Adam Ableman. *The Epistemic Lineage Engine.* Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-E0EC3DC7EABF5060;
source SHA-256
CDD74D9AFAB6532645E68F7F6C9F0F7A9E472D9B8E717290522566FF8DC01F11.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Editorial source notes retained from predecessor

The entries below are corpus-internal comparison and dependency sources. Their inclusion does not establish external validation of this paper's empirical, safety, novelty, or performance claims.

Adam Ableman *The Three Hash Rule.* Zenodo, 2026. doi:10.5281/zenodo.20788617

Adam Ableman *Governed Semantic Operation Language.* Zenodo, 2026. doi:10.5281/zenodo.21214886

Adam Ableman *A Theory of Agentic Observation.* Zenodo, 2026. doi:10.5281/zenodo.20437813

Adam Ableman *CARA: Constrained Artificial Reasoning Architecture.* Zenodo, 2025. doi:10.5281/zenodo.18099715

Adam Ableman *Constitutional Computation.* Zenodo, 2026. doi:10.5281/zenodo.20624488

Adam Ableman *Constraint-Bearing Composability.* Zenodo, 2026. doi:10.5281/zenodo.21632061

Adam Ableman *Fractal Governance Architecture.* Zenodo, 2026. doi:10.5281/zenodo.20754189

Adam Ableman *Governed Semantic Compilation: Lawful Continuation, Refusal Semantics, and Epistemic Promotion in Large Language Models.* Zenodo, 2026. doi:10.5281/zenodo.18453681

Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus

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DOI: 10.5281/zenodo.23847766